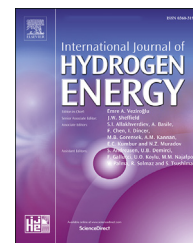


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Experimental study and kinetic modeling of NH_3/CH_4 co-oxidation in a jet-stirred reactor

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HIGHLIGHTS

- NH_3/CH_4 co-oxidation is experimentally studied in JSR.
- Strong competition exists for NH_3 and CH_4 oxidation in lean condition.
- Dual NO formation is observed for NH_3/CH_4 .
- Current existing NH_3 mechanisms need further optimization in low-intermediate temperature range.

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ABSTRACT

This work reports an experimental and kinetic modeling study of the co-oxidation behavior of NH_3/CH_4 binary fuel in a jet-stirred reactor (JSR). Experimental tests are conducted by varying the reaction temperature (850–1350 K), equivalence ratio (0.5–2), and NH_3 mole fraction in fuel (0–100%) individually at ambient pressure and with N_2 being diluent gas. The experimental measurement shows that there exists strong competition between CH_4 and NH_3 oxidation in fuel-lean condition, where the addition of NH_3 tends to suppress CH_4 oxidation by postponing the peak CO formation to a higher reaction temperature. However, such change is not observed in stoichiometric and fuel-rich conditions. When gradually increasing the reaction temperature, a dual NO formation behavior can be noticed for both NH_3/CH_4 mixture and pure NH_3 regardless of equivalence ratio. When gradually increasing the equivalence ratio, NO formation becomes higher with the addition of NH_3 at 1100 K, while is gradually reduced with the addition of NH_3 at 1300 K. Kinetic modeling of the present experimental cases is further performed with different existing reaction mechanisms for NH_3/CH_4 and NH_3 . The comparison of NO emission shows that there is still some space for the improvement of the currently existing reaction mechanisms, especially in the low-intermediate temperature range (850–1200 K) where NO is significantly under-predicted by all models for both NH_3/CH_4 and NH_3 . Further NO reaction pathway and sensitivity analysis suggest that the reactions associated with C–N interaction, H_2NO , NH_2 , N_2H_2 and NNH are responsible for the under-prediction of NO in the low-intermediate temperature range, and their reaction parameters should be optimized for obtaining improved performance.

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Introduction

To mitigate the global warming problem faced by all nations, a consensus has been reached on transition from traditional high-carbon fossil fuels to renewable and low-carbon fuels for future energy supply. In this context, hydrogen, as a carbon-free fuel, has attracted extensive attention since it can be produced through water electrolysis using electricity generated by renewable energy. However, the large-scale use of hydrogen is still facing difficulties in its high cost of storage and transportation [1]. Hence, it is essential to find an alternative carrier of hydrogen, such as ammonia.

In recent years, ammonia has been given a lot of attention owing to its advantages in established safety and economy during storage and transportation, higher volume energy density, and mature production technologies, comparing to liquid hydrogen [2,3]. Worth mentioning is that ammonia has been long attempted as fuel for gas turbines and internal combustion engines since the late 1960s [4,5]. However, the relatively low reactivity of ammonia leads to changes in its basic combustion characteristics, such as slow flame propagation speed [6], high ignition energy [4], and long ignition delay time [7], which together bring some drawbacks to the direct use of ammonia in combustion equipment, like narrow flammable limit [8,9] and weak flame stability [10,11]. More importantly, because of the high N content, the combustion of ammonia also imposes remarkably high NO_x emissions [12]. In order to propose effective strategies to overcome the aforementioned disadvantages of ammonia during its combustion, plenty of studies have been carried out in the past several decades towards a better understanding of its basic combustion characteristics.

Table 1 summarizes the fundamental investigations relevant to ammonia combustion within the recent two decades. The works can be categorized into three main aspects according to their research objectives, namely laminar flame speed (LFS), ignition delay time (IDT), and key product emission (KPE). This information not only helps to deepen the knowledge towards basic combustion characteristics for NH₃ contained fuels, but also play important roles in developing and optimizing chemical reaction kinetic models.

The studies on LFS and IDT in Table 1 have suggested that mixing NH₃ with other fuels (i.e., CH₄, H₂ and DME etc.) can accelerate LFS and shorten IDT for the fuel blends. Specifically, the LFS of NH₃/CH₄ mixture can be accelerated almost linearly by CH₄ addition [6].

The LFS and IDT data from Table 1 has been used for optimizing kinetic models relevant to NH₃ in the past, and kinetic models for different fuels have been proposed with demonstrated reliability mainly in terms of LFS and IDT. However, challenges are always encountered by these kinetic models when predicting KPE from homogeneous or laminar flow reactors in a certain temperature range [13,17,19], as well as NO formation of various flames [44–47], even with the latest ones. In this regard, it is still necessary to make further efforts on the optimization of current kinetic models relevant to NH₃, especially for the prediction of KPE.

On the other hand, from Table 1, it is obvious that a good database has been established for unitary fuel of NH₃ in terms

of LFS, IDT and KPE, while further dedication is required for binary and ternary fuels, especially on KPE, which is even more important for evaluating the performance of the kinetic models. So far, for binary fuel of CH₄ and NH₃, KPE data can be found only in Ref. [24] with JSR (Jet-stirred reactor) and PFR (Plug-flow reactor) apparatuses. However, in Ref. [24] NH₃ only accounted for 4.76% of the fuel volume, and the concentration of NH₃ was not altered. Thus, the conversion behavior of NH₃ in NH₃/CH₄ mixture with high concentration is not well-understood.

This paper intends to provide new experimental KPE data for NH₃/CH₄ mixture by examining the co-oxidation process of NH₃/CH₄ binary fuel in a JSR. Parameters such as reactor temperature, global equivalence ratio, and NH₃ content in fuel mixture are changed individually to investigate their influences on emissions of CO and NO. Furthermore, the present measurement is used to test the performance of various existing kinetic models in predicting the KPE for different NH₃/CH₄ mixtures. By employing the kinetic models with relatively higher reliability in certain temperature ranges, NO formation pathways are analyzed to offer suggestions on future optimization of models for NH₃/CH₄ mixture and pure NH₃, respectively.

Experimental setup

JSR apparatus

NH₃/CH₄ co-oxidation experiment was carried out in a JSR apparatus, which consists of four parts: a reaction chamber made of fused silica, an electrically heated high-temperature tube furnace (HTTF), a gas supply system, and an exhausted gas measuring system, as schematically shown in Fig. 1. To be frank, the catalytic reaction on the reactor wall may have an impact on the consumption of fuel. In specific, Manna et al. [13] found that a reactor made of aluminum oxide is more effective than quartz in promoting the surface catalytic effect. Moreover, given the same material, laminar flow reactor is more likely to facilitate surface catalytic effect than JSR because of its greater surface to volume ratio [17]. For such reasons, JSR made of quartz is considered for study in the current work, in order to alleviate the surface catalytic as far as possible.

The JSR reaction chamber has an inner diameter of 45 mm, resulting in a volume of approximately 47.7 cm³. Fig. 2 displays the geometry structure of the JSR. Specifically, the JSR is designed with two inlets for supplying oxidizer (O₂/N₂) and fuel (NH₃/CH₄) separately, to avoid early reaction prior to their mixing. Oxidizer flows into the left inlet to go through a longer heating duration to guarantee that its temperature is close to the furnace temperature. While fuel flows into the right inlet to avoid thermal decomposition owing to shorter heating duration. An additional pipe is connected to the reaction chamber through the vertical centerline for measuring the JSR inner temperature by inserting a K-type thermocouple (uncertainty of ±5 K).

The HTTF has a maximum heating load of 1.5 kW, and can heat up the furnace temperature up to 1375 K. Tank gases are used as reactant sources with the gas purities higher than

Table 1 – Recent fundamental studies on NH₃ and its mixtures.

Oxidation and pyrolysis behaviors						
Fuel	NH ₃ concentration in reactant	Reactor	Pressure	Equivalence ratio	Temperature	Reference
NH ₃	8%	JSR	1.2 atm	1	900–1350 K	[13]
NH ₃	8%	PFR	1.2 atm	1	1000–1350 K	[13]
NH ₃	2%	JSR	1 atm	0.1–1.0	700–1200 K	[14]
NH ₃	500 ppm	JSR	1 atm	0.0188–0.0093	500–1200 K	[15]
NH ₃	1000 ppm	PFR	1 atm	0.375	1300–2000 K	[15]
NH ₃	2000 ppm	JSR	1 atm	0.6–1.5	800–1300 K	[16]
NH ₃	7.2%–8.6%	JSR	1.2 atm	0.8, 1, 1.2	900–1350 K	[17]
NH ₃	8%	PFR	1 atm	1	900–1350 K	[17]
NH ₃	45.5%–62.5%	PMB	1 atm	0.625–1.25	–	[18]
NH ₃ /NO	500–1000 ppm	JSR	1 atm	0.1–2	1100–1450 K	[19]
NH ₃ /N ₂ O	444.4 ppm	JSR	1.02 atm	–	850–1180 K	[20]
NH ₃ /H ₂	600–2000 ppm	JSR	1 atm	0.25 and 1	800–1280 K	[21]
NH ₃ /H ₂	2000 ppm	JSR	1 atm	0.6–1.5	800–1300 K	[16]
NH ₃ /H ₂	50%–80%	PMB	1 atm	0.8	–	[22]
NH ₃ /H ₂	60%–90%	PMB	1 atm	0.9–1.5	–	[23]
NH ₃ /CH ₄	500 ppm	JSR	1 atm	0.5, 1, 2	600–1200 K,	[24]
NH ₃ /CH ₄	500 ppm	PFR	1 atm	0.5, 1, 2	1200–2000 K	[24]
NH ₃ /CH ₄	30%–80%	PMB	1 atm	0.8	–	[22]
NH ₃ /CH ₄	5%	PMB	1 atm	0.973	–	[25]
Laminar flame speed						
NH ₃	100%	LPF	1 atm	0.6–1.8	413 K	[6]
NH ₃	100%	SPF	1–5 atm	0.7–1.3	298 K	[26]
NH ₃	100%	LPF	1 atm	0.9–1.2	298K	[27]
NH ₃ /H ₂	55%–100%	LPF	1 atm	0.7–1.6	413 K	[6]
NH ₃ /H ₂	70%–100%	SPF	1–10 atm	0.8–1.4	298–473 K	[28]
NH ₃ /H ₂	0–100%	SPF	1–5 atm	1	298 K	[29]
NH ₃ /H ₂	3%–100%	LPF	1–5 atm	0.7–1.6	413 K	[30]
NH ₃ /H ₂	40%–100%	SPF	1 atm	0.8–1.4	298–473 K	[31]
NH ₃ /H ₂	80%–100%	LPF	1–5 atm	0.7–1.4	360K	[32]
NH ₃ /H ₂ /N ₂	0–100%	LPF	1 atm	0.6–1.4	303 K	[33]
NH ₃ /CO	0–100%	LPF	1 atm	0.7–1.7	413 K	[6]
NH ₃ /CO	3%–100%	LPF	1–5 atm	0.7–1.6	413 K	[30]
NH ₃ /CH ₄	0–100%	LPF	1 atm	0.7–1.5	413 K	[6]
NH ₃ /CH ₄	0–56%	SPF	1 atm	0.8–1.3	298 K	[34]
NH ₃ /CH ₄	0–56%	SPF	1–5 atm	0.7–1.3	298	[35]
NH ₃ /CH ₄	0–80%	SPF	1 atm	0.9–1.2	298 K	[36]
NH ₃ /CH ₄	0–20%	LPF	5 atm	0.9	298K	[37]
NH ₃ /syngas	20%–100%	LPF	1–5 atm	0.7–1.6	413 K	[30]
Ignition delay time						
NH ₃	100%	ST	1.4–30 atm	0.5–2	1560–2455 K	[38]
NH ₃	100%	ST	20–40 atm	0.5–2	1100–1600 K	[7]
NH ₃ /H ₂	80%–100%	RCM	20–60 atm	0.5–2	950–1150K	[39]
NH ₃ /H ₂	90%–100%	RCM	20–75 atm	0.5–3	1040–1210	[40]
NH ₃ /H ₂	5.31%–12.29%	RCM	43–65 atm	0.2, 0.35, 0.5	1000–1100 K	[41]
NH ₃ /CH ₄	50%–100%	RCM	20–70 atm	0.5–2	930–1140 K	[42]
NH ₃ /DME	0–18.2%	RCM	10–70 atm	0.5–2	610–1180 K	[43]

*JSR: Jet-stirred reactor; PFR: Plug-flow reactor; LPF: Laminar premixed flame; SPF: Spherical premixed flame; PMB: Porous media burner; ST: Shock tube; RCM: Rapid compression machine.

99.9%, and pressure valves are used to adjust the pressure inside the JSR to be fixed at approximately 0.1 MPa. The gas flowrates are monitored and changed by electrical mass flowmeter controllers (Seven-star CS200A, precision of $\pm 0.35\%$ full scale). A portable Kane KM950 gas analyzer is used to measure the components of the exhausted gas (O₂, NO and CO) with a relative error of $\pm 5\%$ reading. It is also necessary to note that, the exhausted gas is cooled down to room temperature and dried by silica-gel before entering the gas analyzer, hence the gas concentrations are in a dry basis for all tests.

Due to heat loss at the two ends of the HTTF, the temperature is not ideally constant along the furnace centerline. To verify the temperature homogeneity inside the HTTF, furnace temperature (T_{fur}) is sampled with a K-type thermocouple along the furnace axis at an interval of 10 mm and under three different targeted heating temperatures (T_{set}). Fig. 3 plots the measured T_{fur} profiles for T_{set} being 1173 K, 1273 K and 1373 K. It is clear that T_{fur} is below T_{set} near the two ends, while exceeds T_{set} in the middle of the furnace. Nevertheless, a fairly good agreement between T_{fur} and T_{set} is obtained at a distance of about 50–60 mm from the entrance of the furnace

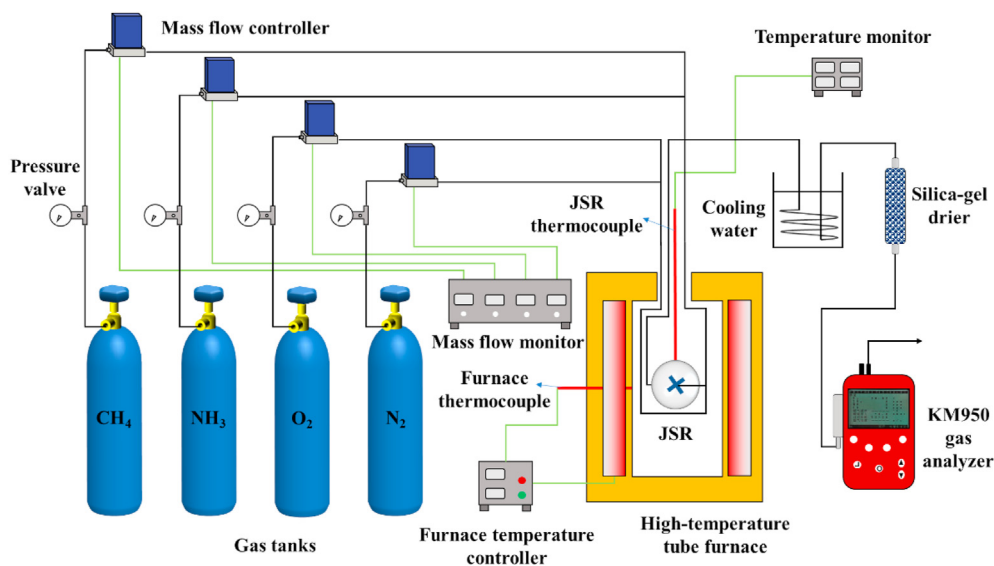


Fig. 1 – Schematic diagram of JSR apparatus.

regardless of T_{set} . Therefore, the JSR is placed inside the HTTF with its center fixed at 55 mm from the entrance.

Inside the HTTF, JSR receives radiative heat from the furnace wall, and transfers heat to the inner reactants. Considering the heat transfer process, it is necessary to check the consistency of temperature between T_{set} and T_{JSR} (temperature in the JSR center). Fig. 4a draws the measured T_{JSR} scatter plot against T_{set} from 800 K to 1350 K. The red dashed line represents the idea curve where T_{JSR} equals to T_{set} . It can be seen that the scatter points are located basically on the curve, suggesting a good match between T_{JSR} and T_{set} .

Since JSR is regarded as a zero-dimensional reactor due to its homogeneous assumption, it is also necessary to check the temperature uniformity inside the JSR. To this end, the temperature throughout the JSR vertical diameter is measured at four different targeted furnace temperatures from 900 K to 1200 K. Note that, at this stage, T_{JSR} is measured in reacting condition (stoichiometric CH_4/O_2 mixture with O_2 mole fractions being 0.5%, N_2 being diluent, and total flowrate being 1 L/min). Fig. 4b shows the temperature distributions inside the

JSR under different T_{set} conditions. It can be seen that T_{JSR} is generally higher than T_{set} due to chemical reaction, but the discrepancy is within 20 K in most regions, which corresponds to a maximum error of 2.2% when T_{JSR} equals to 900 K. Nevertheless, temperature fluctuation can be hardly observed inside the JSR. In this regard, the currently designed JSR is considered to be valid for the zero-dimensional assumption.

Table 2 specifies the operating conditions in the present experimental study. Specifically, the JSR is operated under atmosphere pressure with the total flow rate being fixed as 1L/min at room temperature (298 K). To minimize the influence of chemical reaction on reactor temperature, the total flow rate of fuel (mixture of CH_4 and NH_3) is maintained as low as 5 mL/min (corresponding to 5000 ppm, 0.5% in volume) in all cases. The volume fractions of NH_3 in fuel (X_{NH_3}) are changed from 0% to 100% at an interval of 20%. N_2 is used as the balance gas. Overall equivalence ratio (Φ) is defined on the consideration of CO_2 , H_2O and NO as final oxidation products for the fuel blends according to Arunthanayothin et al. [24], since the

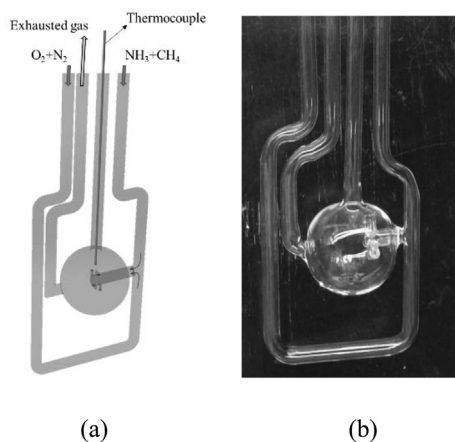


Fig. 2 – (a) Schematic diagram and (b) physical picture of JSR.

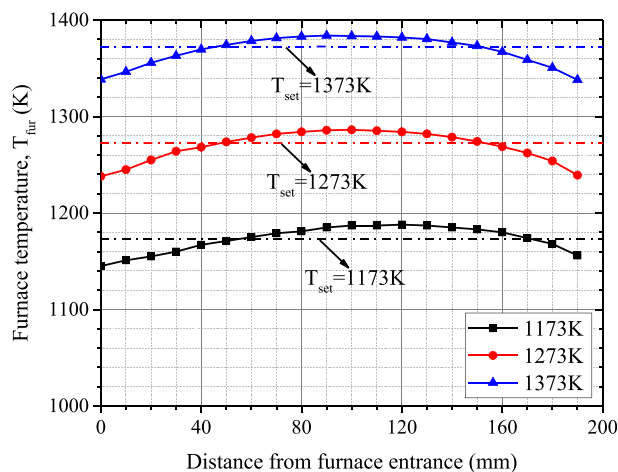


Fig. 3 – Measured T_{fur} along the furnace centerline in different T_{set} .

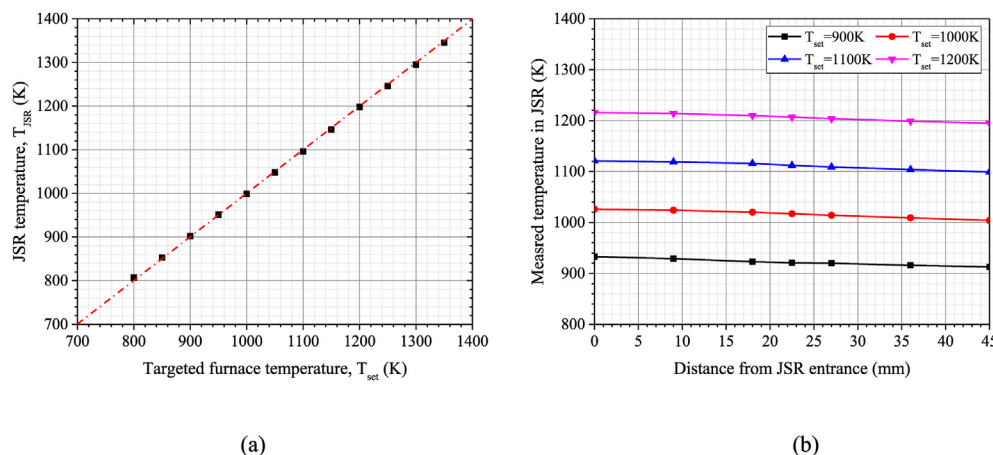


Fig. 4 – (a) Temperature comparison between JSR and targeted value and (b) measured temperature profiles along JSR vertical diameter.

formed NO amount can not be neglected in most conditions. Φ is varied from lean condition to rich condition by fixing the flowrates of fuel and N_2 while adjusting the amount of the supplied O_2 . The JSR is operated in a furnace temperature range of 850 K–1350 K. The actual residence time of reactants is calculated as the ratio of JSR volume to reactants volume flow rate at the corresponding temperature. Given the above information regarding JSR and reactants flow rate, the actual residence time (τ) is a function of T_{JS} , and is calculated by $894/T_{JS}$ in the present work. Moreover, due to the small surface/volume ratio of the current JSR, the influence of heterogeneous reactions at reactor inner surface on the fuel oxidation behavior can be neglected according to previous work [48]. For all the experimental cases listed in Table 2, each one has been repeated for three times to verify the reproducibility of the measurement.

Verification of experimental procedure

To verify the reliability of the present experimental procedure, the experimental case of $CH_4/O_2/N_2$ oxidation in JSR

performed by Li et al. [49] is reproduced. In particular, Li et al. [49] carried out $CH_4/O_2/N_2$ oxidation in the temperature range of 800K–1350K with a fixed CH_4 concentration of 5000 ppm, in a fuel-lean ($\Phi = 0.5$) and a fuel-rich ($\Phi = 1.43$) condition, respectively. The reactor volume of was 60 cm^3 , and the total flow rate of the mixture was fixed at 1 L/min. For the validation purpose, the total flow rate in the present work is decreased to 0.8 L/min to match the identical residence time.

Fig. 5a and b plot the O_2 and CO mole concentration profiles as a function of reactor temperature in lean and rich conditions, respectively. In lean condition, O_2 begins to be consumed at the temperature of 1050 K, where CO production reaches its peak according to Li's experimental data [49]. Such behavior is well-reproduced by the present experiment. In rich condition, the onset of O_2 consumption is postponed to 1200 K according to Li's experiment, however, it is slightly earlier in the present experiment. For CO profile, the rise and drop trend after 1200 K is exhibited by both experiments, while the peak location is shifted to the higher temperature region by approximately 50 K, and the peak value is around 1000 ppm lower in the present experiment. Such discrepancy is probably caused by the different designs of the reactor

Table 2 – Experimental conditions for the considered cases.

Case	CH_4 (ppm)	NH_3 (ppm)	O_2 (vol%)	τ	Φ	T (K)
1	5000	0	2, 1, 0.7	$894/T$	0.5, 1, 1.5	850–1350
2	4000	1000	1.9, 0.9, 0.6	$894/T$	0.5, 1, 1.5	850–1350
3	3000	2000	1.7, 0.9, 0.6	$894/T$	0.5, 1, 1.5	850–1350
4	2000	3000	1.6, 0.8, 0.5	$894/T$	0.5, 1, 1.5	850–1350
5	1000	4000	1.4, 0.7, 0.5	$894/T$	0.5, 1, 1.5	850–1350
6	0	5000	1.3, 0.6, 0.4	$894/T$	0.5, 1, 1.5	850–1350
7	0	5000	1.3–0.3	0.813	0.5–2	1100
8	0	5000	1.3–0.3	0.745	0.5–2	1200
9	0	5000	1.3–0.3	0.688	0.5–2	1300
10	2000	3000	1.6–0.4	0.813	0.5–2	1100
11	2000	3000	1.6–0.4	0.745	0.5–2	1200
12	2000	3000	1.6–0.4	0.688	0.5–2	1300
13	4000	1000	1.9–0.5	0.813	0.5–2	1100
14	4000	1000	1.9–0.5	0.745	0.5–2	1200
15	4000	1000	1.9–0.5	0.688	0.5–2	1300

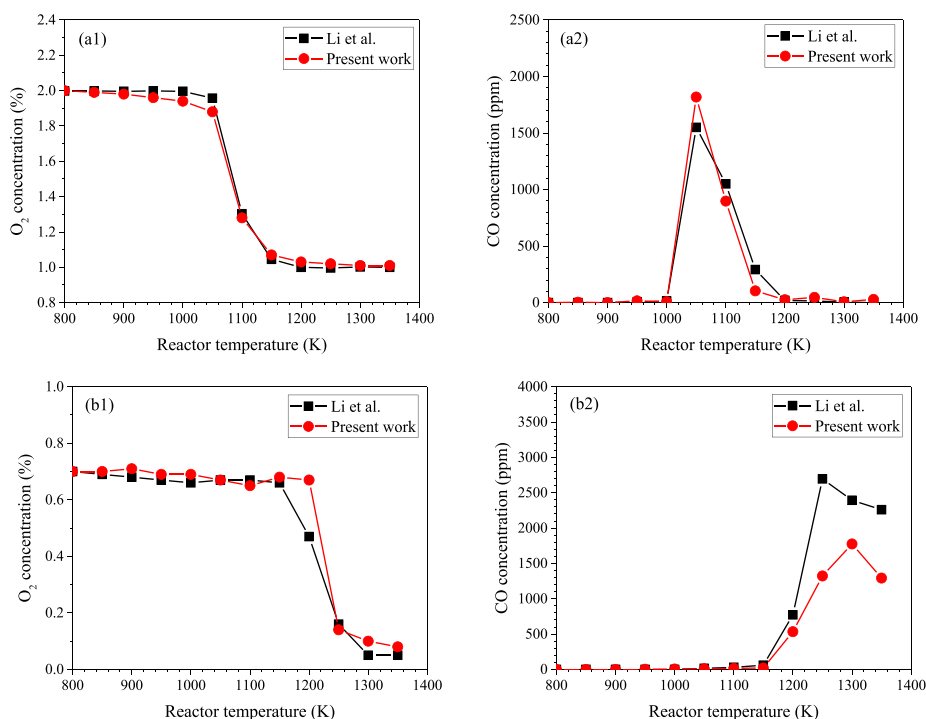


Fig. 5 – Experimental reproduction results of Li's work [49] in (a) lean condition ($\Phi = 0.5$) and (b) rich condition ($\Phi = 1.43$).

inlets. In the experiment of Li et al. [49], the oxidizer and fuel are premixed before entering the reactor, thus earlier oxidation of fuel might take place, and the actual reaction volume is expected to be larger than the reactor volume. A kinetic modeling test using a larger reactor volume indeed shows an earlier production of CO peak (not shown here), which confirms such speculation. Nevertheless, the present validation results agree with Li's experimental data generally well, thus the experimental procedure is regarded to be reliable.

Results and discussion

Effect of reaction temperature

CH₄/NH₃ co-oxidation experiment is firstly conducted under three equivalence ratios (0.5, 1 and 1.5) in the reaction temperature range of 850 K–1350 K for mixtures with different X_{NH_3} . The detailed operational information for this set of experiments is shown in Cases 1–6 of Table 2.

Fig. 6 plots the CO profiles as a function of reaction temperature for various X_{NH_3} at three equivalence ratios, namely 0.5, 1 and 1.5. Since CO formation is avoided in the pure NH₃ ($X_{\text{NH}_3} = 100\%$) oxidation case, the CO emission profile is not shown for this case.

In lean condition ($\Phi = 0.5$), CO firstly increases and then decreases no matter what X_{NH_3} is. More importantly, the maximum CO concentration is found to reduce as X_{NH_3} is enlarged, and the temperature for the production of peak CO is gradually boosted when X_{NH_3} exceeds 40%. For the lower X_{NH_3} cases (<40%), the temperature for maximum CO is observed to be unchanged, which is considered to be due to

the large interval of reactor temperature (50 K). Nevertheless, it follows that the addition of NH₃ in CH₄ will not only reduce the maximum CO generation, but also elevate the onset temperature for the maximum CO concentration, especially in high NH₃ blending ratio scenarios. This phenomenon indicates that the addition of NH₃ promotes the reaction temperature of NH₃/CH₄ mixture in lean condition. Due to the fact that NH₃ has higher reaction energy than CH₄, the reaction temperature of the CH₄/NH₃ mixtures increase when X_{NH_3} exceeds 40%.

In stoichiometric condition ($\Phi = 1$), the same trend where peak CO concentration minimizes with the increase in X_{NH_3} is obtained as that in lean condition, however the temperature for producing the maximum CO is found to be independent of X_{NH_3} . Such phenomenon is also noticed in rich condition ($\Phi = 1.5$). On the other hand, comparing to lean condition, the temperature for peak CO generation is found to be raised to 1200 K in stoichiometric condition and 1300 K in rich condition, respectively. It then derives that the onset of CH₄ oxidation tends to require a higher temperature when enlarging the equivalence ratio.

Fig. 7 draws the NO profiles as a function of reaction temperature for various X_{NH_3} at the three equivalence ratios. It is interesting to notice a dual-staged NO formation behavior for all fuel blends regardless of equivalence ratio. Specifically, the first NO formation stage occurs at a lower temperature before 1200 K, while the second NO formation stage occurs generally above 1300 K and shows a monotonous increasing trend with reaction temperature.

In the first NO formation stage in lean condition, the addition of NH₃ on one hand elevates the NO peak concentration as an expected result, on the other hand gradually

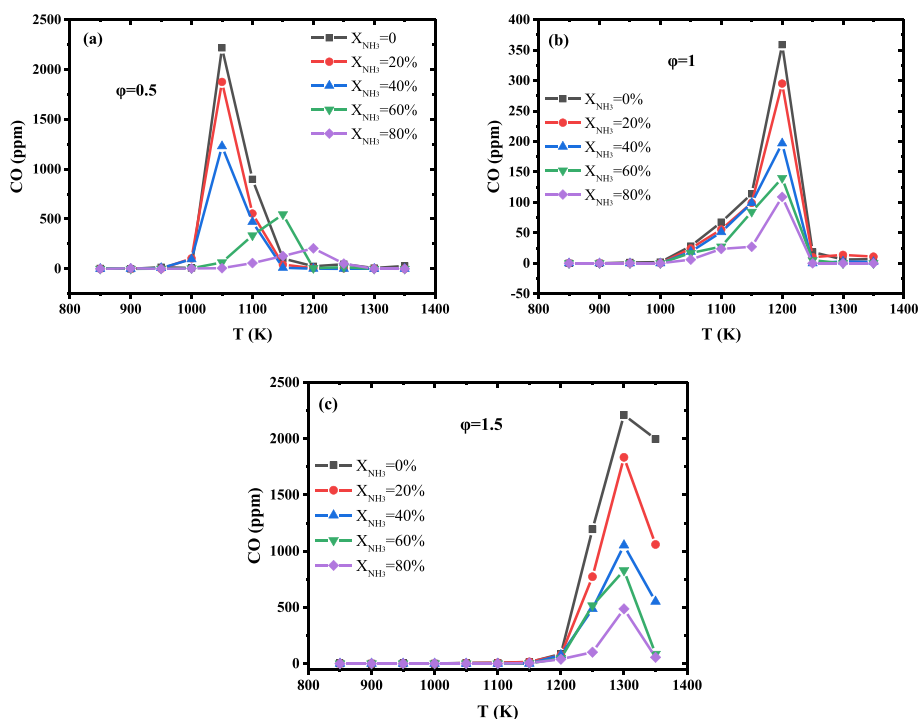


Fig. 6 – CO profiles as a function of reaction temperature for various CH_4/NH_3 mixture under different equivalence ratios.

pushes the temperature for peak NO generation towards the higher temperature direction. Interestingly, a competition between the oxidation of CH_4 and NH_3 can be noticed by combining Figs. 7a and 6a in lean condition. Specifically, for NH_3/CH_4 mixture with X_{NH_3} equals to 80%, the peak NO occurs at 1050 K, while the peak CO occurs at 1200 K. In the second stage of NO formation, the inflection points of the NO curves

where the lowest NO emissions are obtained are postponed with the addition of NH_3 . As a result, at an identical temperature above 1250 K, the enhancement of X_{NH_3} tends to generate a lower NO emission.

In stoichiometric condition, the peak NO concentration in the first NO formation stage gradually increases as X_{NH_3} is enlarged, while the temperature for such NO peak is seen to be

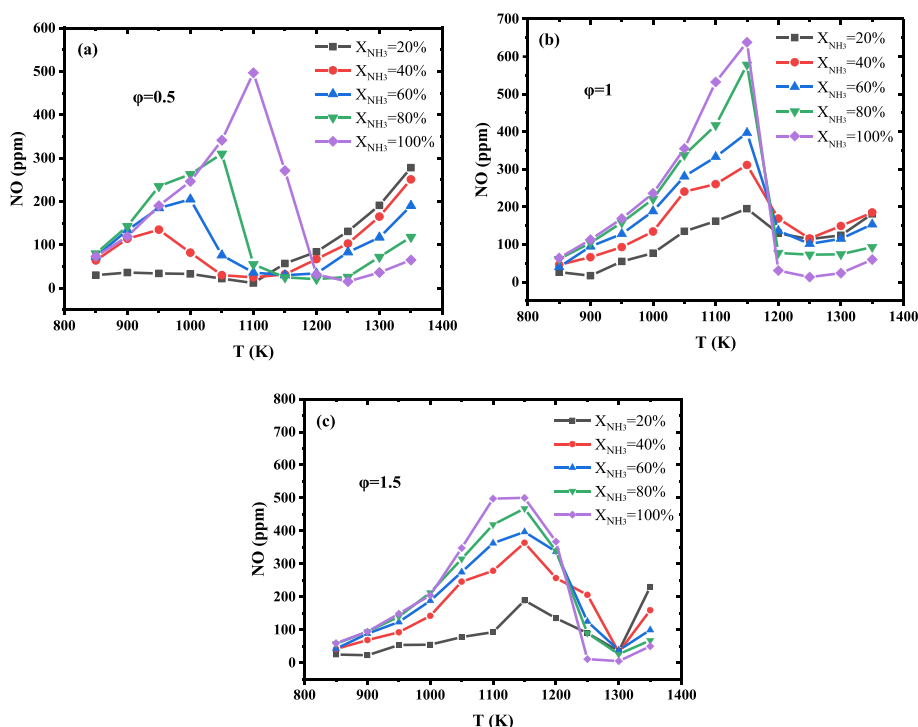


Fig. 7 – NO profiles as a function of reaction temperature for various CH_4/NH_3 mixture under different equivalence ratios.

nearly fixed at 1150 K regardless of X_{NH_3} . In the second NO formation stage where reaction temperature exceeds 1250 K, NO concentration continually increases with reaction temperature, and the inflection point is basically fixed at 1250 K for all fuel blends. Interestingly, the inflection point is further boosted to 1300 K in rich condition (see Fig. 7c).

For the current reaction temperature range, NO is not likely to be formed via thermal-route. Hence, the only reason for the production of NO in low-intermediate temperature range (850 K–1200 K) is due to the oxidation of NH_3 . It can be then deduced that there exists a competition between CH_4 and NH_3 oxidation in the current reaction system. The competition not only affects the peak concentrations of CO and NO but also affects the temperatures for generating the highest as well as lowest NO emissions. Such interaction has been also found for NH_3/H_2 mixture during its oxidation in JSR [16].

Fig. 8 shows the O_2 profiles as a function of reaction temperature for various X_{NH_3} at the three equivalence ratios. It can be seen that O_2 begins to decrease at a lower temperature in lean condition comparing to stoichiometric and rich conditions. Specifically, in lean condition, O_2 starts to deplete at 1100 K for all fuel blends, and its level keeps constant when reaction temperature exceeds 1250 K, except the pure NH_3 case ($X_{\text{NH}_3} = 100\%$). In addition, the temperature for rapid consumption of O_2 is in agreement with the temperature for peak CO and NO production (see Figs. 6a and 7a).

In stoichiometric condition, O_2 begins to be significantly consumed at 1200 K, and such observation also applies in rich condition. The phenomenon illustrates that the oxidation of fuel (indicated by the consumption of O_2) tends to take place at a higher reaction temperature, which is in line with the observation in Fig. 6.

Effect of equivalence ratio

CH_4/NH_3 co-oxidation experiment is subsequently performed under three reaction temperatures (1100 K, 1200 K and 1300 K) with equivalence ratio varying from 0.5 to 2.0 for mixtures with different X_{NH_3} . The detailed operational information for this set of experiment is shown in Cases 7–15 of Table 2.

Fig. 9 plots the CO profiles as a function of equivalence ratio for various X_{NH_3} at three reaction temperatures, namely 1100 K, 1200 K and 1300 K. Since CO is eliminated in pure NH_3 oxidation, the CO profiles with X_{NH_3} being 100% are removed from the figures. It can be seen that CO formation has a strong dependence on the reaction temperature. Particularly, CO is mainly formed in fuel lean conditions at 1100 K, while in fuel rich conditions at 1300 K. Moreover, equivalence ratio shows a completely opposite influence on the CO formation between these two reaction temperatures. Interestingly, at the medium temperature of 1200 K, the CO profiles exhibit a peak in stoichiometric condition for mixtures with X_{NH_3} below 60%. For the case with X_{NH_3} being 80%, CO formation continually decreases as the increase of equivalence ratio. Consequently, CO formation is significantly higher in this case than that of other lower X_{NH_3} cases when the equivalence ratio is smaller than 0.8, while CO formation becomes the lowest among all fuel blends of interest when the equivalence ratio exceeds nearly 0.9. The phenomenon indicates that when the NH_3 content exceeds 80%, the formation of CO is promoted at an equivalence ratio below 0.8 at 1200 K, and CO gradually decreases with the increase of equivalence ratio.

Fig. 10 shows the NO formation profiles as a function of equivalence ratio for CH_4/NH_3 mixture with different X_{NH_3} at reaction temperatures of 1100 K, 1200 K and 1300 K,

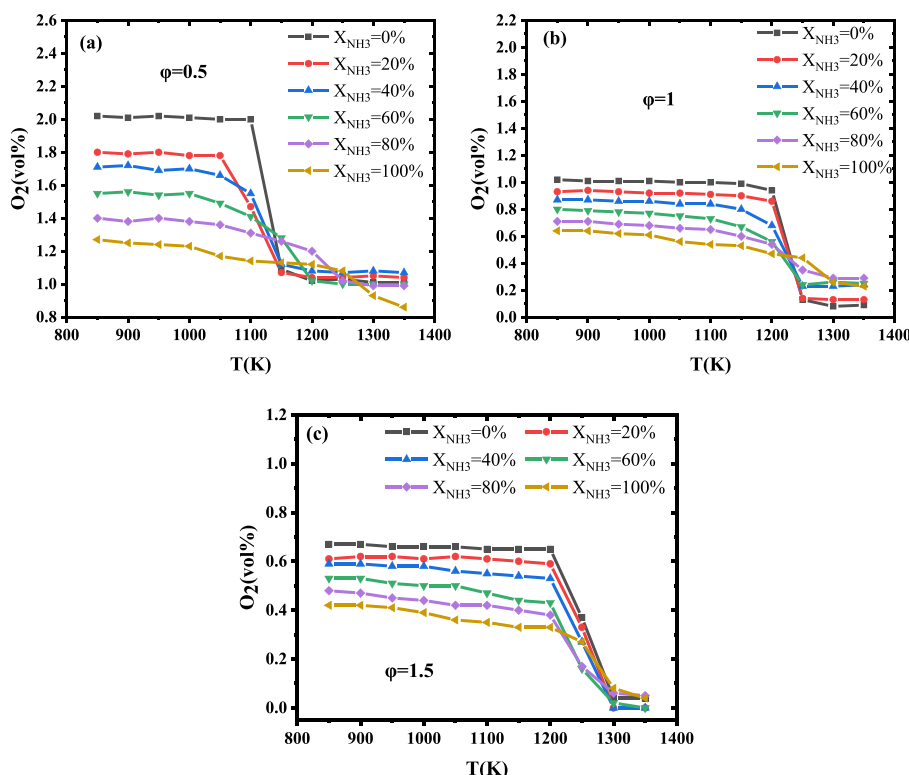


Fig. 8 – O_2 profiles as a function of reaction temperature for various CH_4/NH_3 mixture under different equivalence ratios.

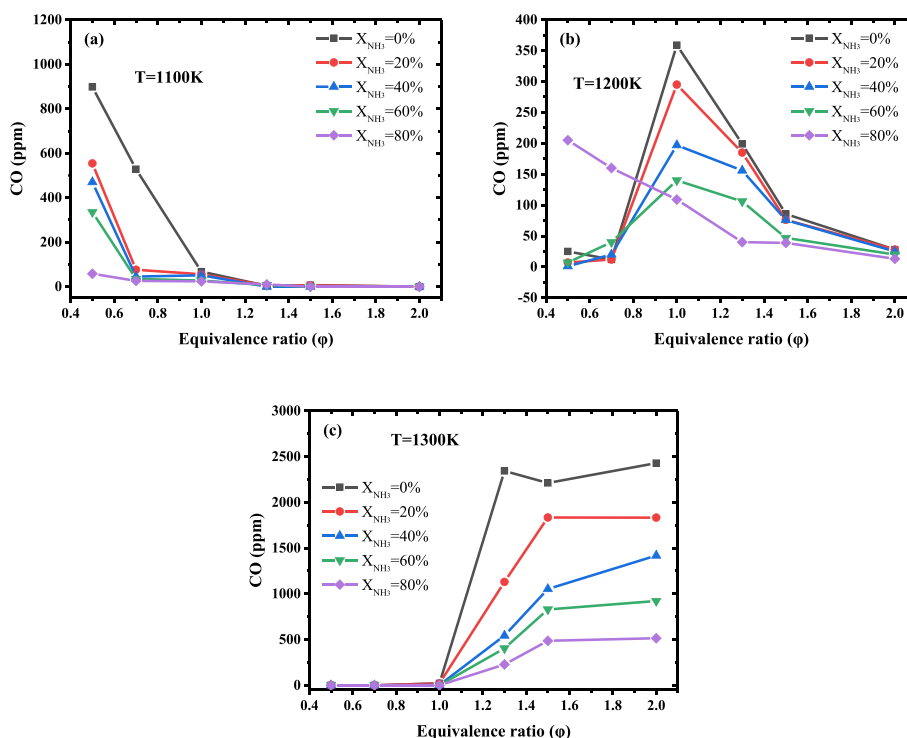


Fig. 9 – CO profiles as a function of equivalence ratio for various CH_4/NH_3 mixture under different reaction temperatures.

respectively. At 1100 K, NO formation generally shows a rising trend then falling trend with the increase of equivalence ratio for all mixtures. Moreover, the NO concentration is found to be higher for larger X_{NH_3} cases. Contrarily, at a higher temperature of 1300 K, NO concentration appears to be reduced with NH_3 addition at an identical equivalence ratio, and an overall

decreasing trend of NO against equivalence ratio is observed for all fuel blends. Such opposite NO formation behavior in respect to that at 1100 K is considered to be caused by the thermal De-NOx mechanism [50]. It is also interesting to notice a totally different NO formation behavior in the medium temperature of 1200 K. Specifically, in lean conditions

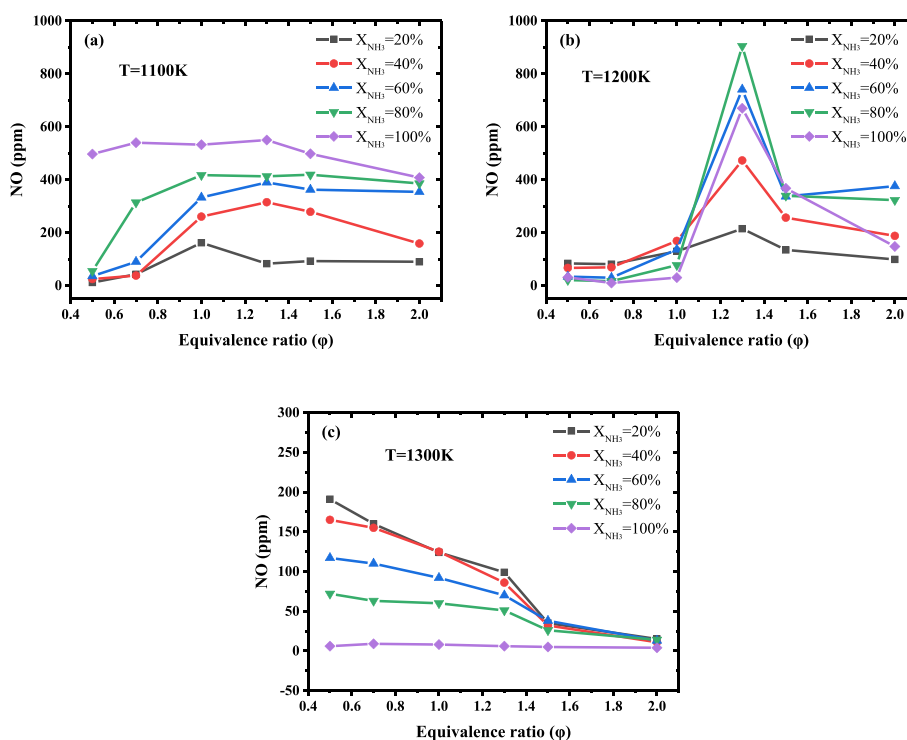


Fig. 10 – NO profiles as a function of equivalence ratio for various CH_4/NH_3 mixture under different reaction temperatures.

($\Phi < 1$), NO formation is not apparently affected by equivalence ratio for each fuel, but NO concentration is generally lower at a higher X_{NH_3} . In rich conditions, all NO profiles exhibit a peak at equivalence ratio of 1.3, the NO is more likely to be produced under rich combustion conditions regardless of X_{NH_3} at 1200K. Mixing CH_4 with NH_3 does not change the equivalent ratio for obtaining the maximum NO concentration. Besides, the peak NO concentration firstly increases with X_{NH_3} in the range of 20% and 80%, and then decreases when X_{NH_3} is further raised to 100%.

Fig. 11 plots the O_2 profiles as a function of equivalence ratio for various X_{NH_3} at three reaction temperatures. At 1100K, O_2 concentrations keep decreasing against the equivalence ratio for all fuel blends. At 1200 K, a continuous decreasing trend of O_2 concentration is found for fuel blends with X_{NH_3} larger than 60%, while saltation is noticed when the equivalence ratio varies between 0.7 and 1.0 for other lower X_{NH_3} cases. At a higher temperature of 1300K, the continuous decreasing trend of O_2 occurs again. The irregular O_2 plots at 1200 K is mainly resulted from the competition of oxidation between NH_3 and CH_4 , as can be clearly noticed from Fig. 8a.

Evaluation of existing detailed reaction mechanisms

CH_4/NH_3 reaction mechanisms

In this section, the performance of previously proposed reaction mechanisms for CH_4/NH_3 mixture on describing the co-oxidation process of CH_4/NH_3 mixture is evaluated. Table 3 summarizes the most recent reaction mechanisms considered for evaluation purpose in this section. Note that, although there are some other reaction mechanisms proposed

Table 3 – Detailed reaction mechanisms to be evaluated for CH_4/NH_3 mixture.

Mechanism ID	Number of species	Number of reactions	Reference
PG2018	151	1397	[50]
CEU2021	91	444	[52]
Tian2009	84	703	[53]
Mendiara2009	97	779	[54]
Li2020	151	1408	[49]
Okafor2018	59	356	[34]

more recently, such as by Arunthanayothin et al. [24] and Sun et al. [51], issues or failures would be presented when incorporating them in JSR kinetic modeling. Hence, they are not included in Table 3.

The evaluation is conducted by using a well-stirred reactor (WSR) model of Chemkin Pro (version 2020 R1) software. The mixture with X_{NH_3} being 40% is taken as a representative fuel. Kinetic modeling is carried out for three equivalence ratios ($\Phi = 0.5, 1, 1.5$) and in the temperature range of 850 K–1350 K, which are in consistency with the current experimental conditions.

Fig. 12 shows the predicted CO concentration profiles against experimental measurement as a function of reaction temperature in the three equivalence ratios. In lean condition (see Fig. 12a), the CO formation behavior is well-reproduced by the mechanisms of PG2018 and Li2020. The mechanisms of Tian2009 and Mendiara2009 predict an earlier presence of CO peak but a good peak value, whereas the mechanisms of Okafor2018 and CEU2021 forecast a noticeably higher reaction temperature for generation CO peak by approximately 50 K

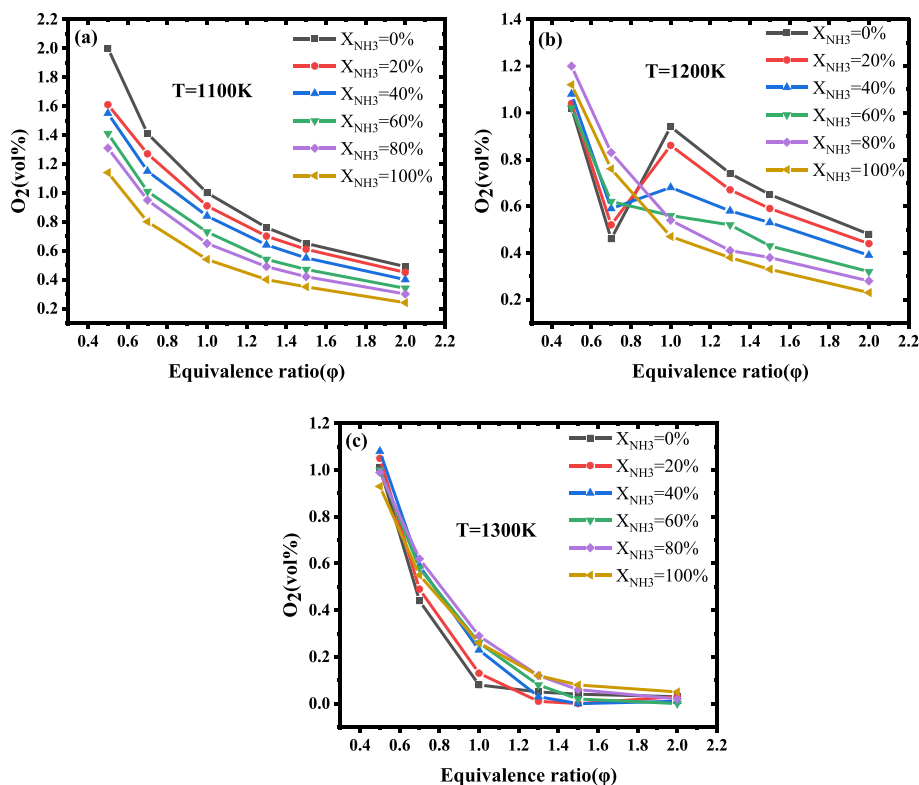


Fig. 11 – O_2 profiles as a function of equivalence ratio for various CH_4/NH_3 mixture under different reaction temperatures.

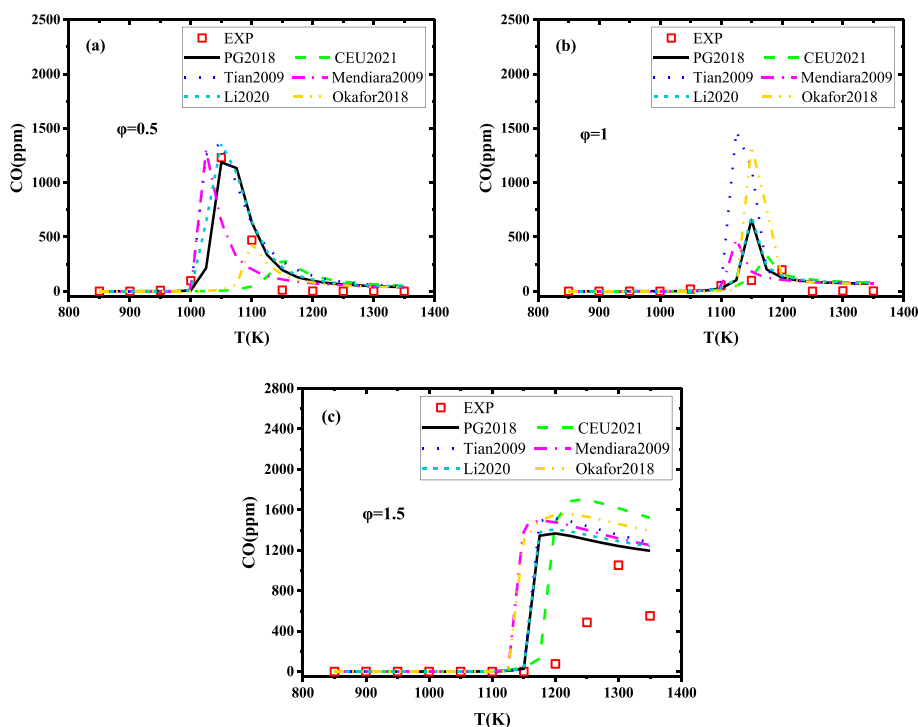


Fig. 12 – Comparisons of CO between experimental measurement and kinetic modeling at three equivalence ratios.

and 100 K, respectively. In stoichiometric condition (see Fig. 12b), the reaction temperature for the peak CO location is under-predicted by all mechanisms, while the CEU2021 mechanism shows the smallest difference. Moreover, the CEU2021 mechanism also exhibits the smallest discrepancy in

the peak CO concentration, thus resulting in the best predictions among all mechanisms. In rich condition (see Fig. 12c), all the considered mechanisms give quite close predictions of CO formation, however, they all fail to reproduce the peak CO temperature location with a difference over

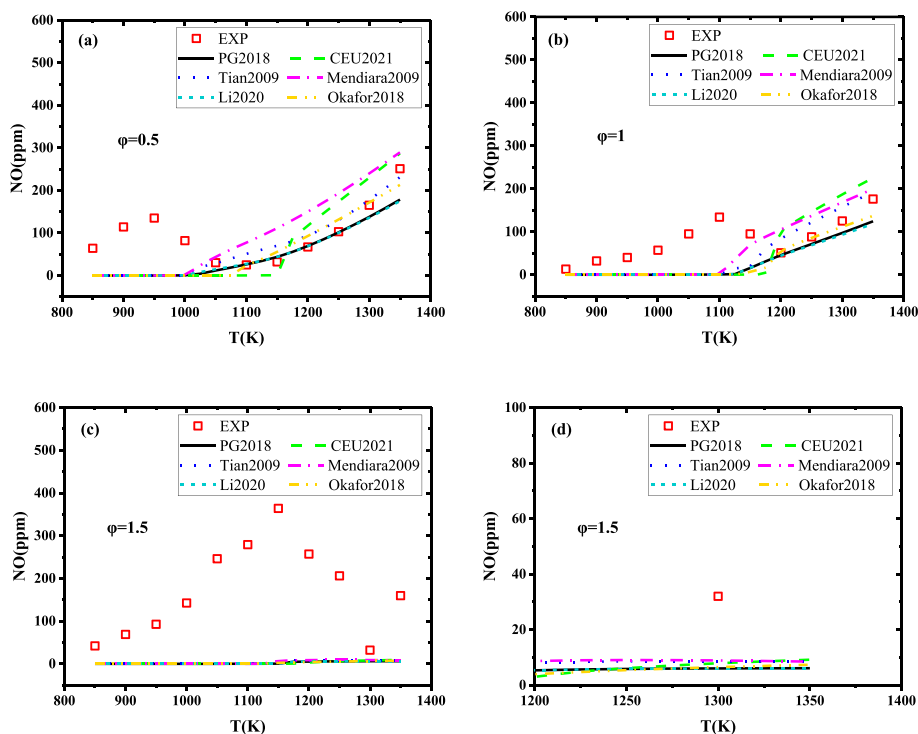


Fig. 13 – Comparisons of NO between experimental measurement and kinetic modeling at three equivalence ratios.

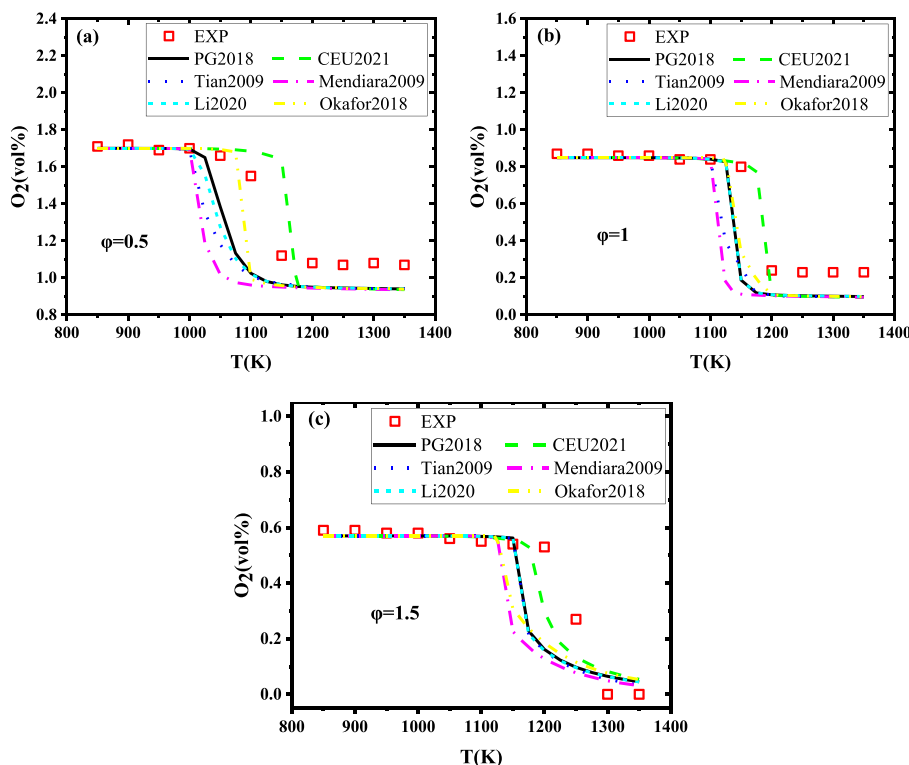


Fig. 14 – Comparisons of O_2 between experimental measurement and kinetic modeling at three equivalence ratios.

100 K. Among them, the mechanism of CEU2021 shows a relatively smaller error with comparison to the measurement for temperature location where peak CO is presented.

Fig. 13 shows the comparison between prediction and experimental measurement for NO formation profiles as a function of reaction temperature in the three equivalence ratios. In lean condition (see Fig. 13a), all the considered mechanisms exhibit high potential of capturing the NO formation behavior in high reaction temperature above 1100K, while remarkably underestimate NO formation in low reaction temperature before 1000 K. Among them, the mechanisms of PG2018, Li2020 and Okafor2018 show a relatively better performance on NO prediction against measured data. In stoichiometric condition (see Fig. 13b), the same performance as in lean condition is observed for all the mechanisms, while the best fitting temperature range moves to a higher starting point of 1200 K. In rich condition (see Fig. 13c), NO is under-predicted for basically the whole temperature range by all mechanisms. Fig. 11d shows a partially enlarged drawing of Fig. 11c in the temperature range of 1200 K–1400 K for better resolution. It is found that the predicted NO concentration is basically less than 10 ppm over the whole temperature range for anyone of the mechanisms, while the lowest measured NO is around 30 ppm at 1300 K. This demonstrates that the currently existing CH_4/NH_3 mechanisms fail to reproduce NO formation not only in low reaction temperature but also in rich condition.

Fig. 14 shows the prediction and experimental measurement of O_2 profiles as a function of reaction temperature in three equivalence ratios. In lean condition (see Fig. 12a), all the considered mechanisms have a good prediction for O_2 below

1000K, and they all indicate the non-reacting process at such low temperature. When reaction temperature exceeds 1000 K, the mechanisms of Mendiara2009, Tian2009, Li2020, Okafor2018 and PG2018 show an earlier consumption of O_2 than experimental data, while CUE2021 gives a delayed result. Nevertheless, CUE2021 is found to show a relatively better prediction of O_2 , especially in stoichiometric and rich conditions.

NH_3 reaction mechanisms

This section further assesses the performance of some recent mechanisms for describing the oxidation process of pure NH_3 without the presence of CH_4 . Table 4 summarizes the reaction mechanisms to be considered for modeling and evaluation purposes. Note that, these mechanisms are mainly proposed for the target of NH_3 and NH_3/H_2 , such that C–N interaction is not contained.

Fig. 15 gives the comparisons of NO formation between experimental measurement and numerical predictions for

Table 4 – Detailed reaction mechanisms to be evaluated for pure NH_3 .

Mechanism ID	Number of species	Number of reactions	Reference
Mathieu 2015	56	278	[38]
Nakamura2017	38	232	[55]
Otomo2018	32	213	[56]
Song2016	33	204	[57]
Tang 2022	35	211	[14]
Xiao 2017	29	108	[46]

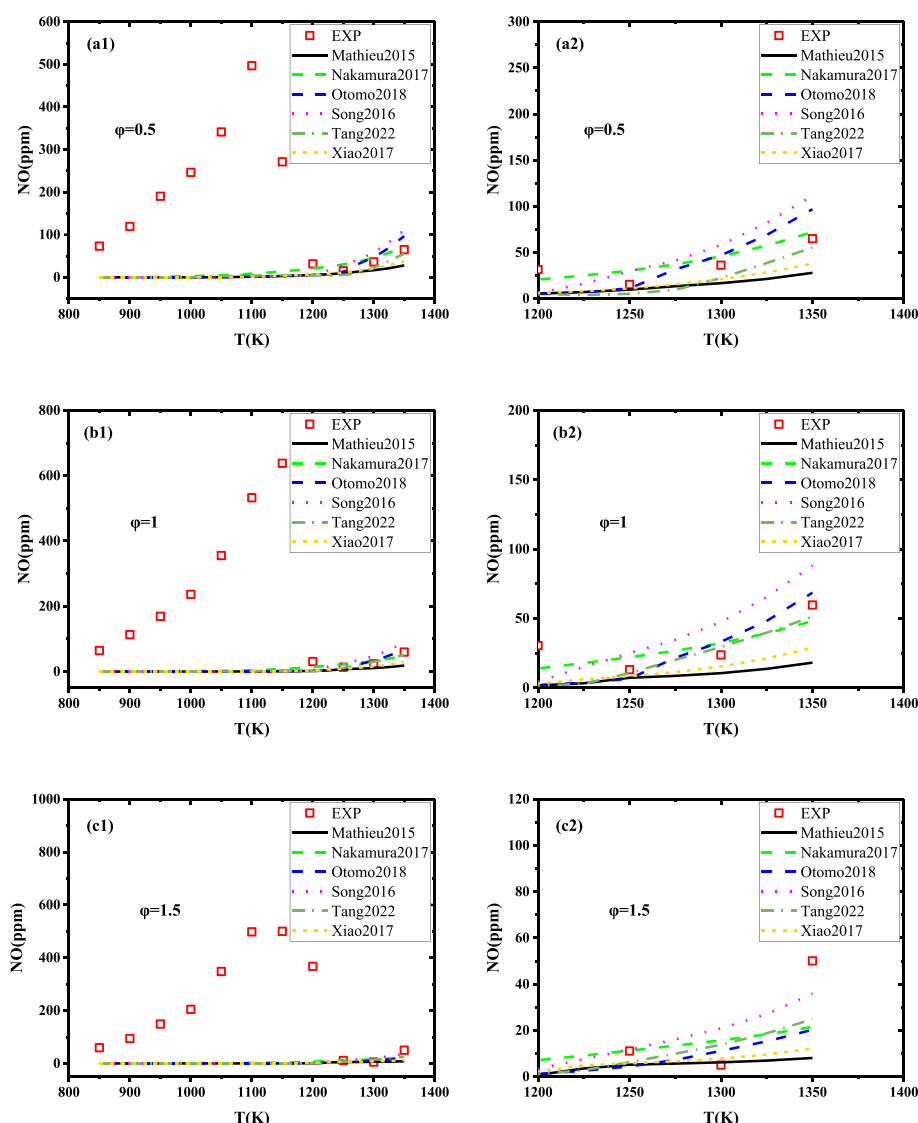


Fig. 15 – Comparisons of NO concentration between experimental measurement and kinetic modeling for pure NH_3 oxidation case in three equivalence ratios.

pure NH_3 oxidation case in three different equivalence ratios. The left column (Fig. 15a1 - 15c1) shows the comparison results in the whole temperature range of interest, while the right column (Fig. 15a2 - 15c2) focuses on the high temperature condition (>1200 K) to enhance the resolution.

According to the left column, all the NH_3 mechanisms considered in this section still fail to capture the NO formation in the temperature range of 850–1200 K. Such huge inconsistency between modeling and experiment has been also reported before by Sabia et al. [17], who investigated pure NH_3 oxidation in a JSR. The discrepancy was claimed to be caused by the poor NH_2 chemistry in low-intermediate temperature range (<1200 K). Nevertheless, it can be still recognized that the mechanism of Nakamura2017 exhibits an earlier production of NO at intermediate temperature, for example 1100 K, with comparison to others.

According to the right column, the measured NO formation in high temperature condition is much better reproduced by all mechanisms regardless of equivalence ratios. Among these

mechanisms, Nakamura2017 gives the best NO prediction in lean and stoichiometric conditions, while Song2016 shows the best performance in rich condition.

Fig. 16 gives the comparisons of O_2 concentration between experimental measurement and numerical predictions for pure NH_3 oxidation case in three different equivalence ratios. According to the experimental measurement, O_2 is weakly consumed from 1000 K to 1200 K. However, none of the considered mechanism, except Nakamura2017, captures such change. Nevertheless, Nakamura2017 strongly under-predicts O_2 at high temperatures.

NH_3 reaction pathway analysis

To reveal the different NO formation processes during NH_3 oxidation with and without CH_4 presence, NH_3 reaction pathway analysis is carried out for NH_3/CH_4 mixture ($X_{\text{NH}_3} = 40\%$) and pure NH_3 , respectively, in stoichiometric and ambient pressure condition. The reaction mechanism of

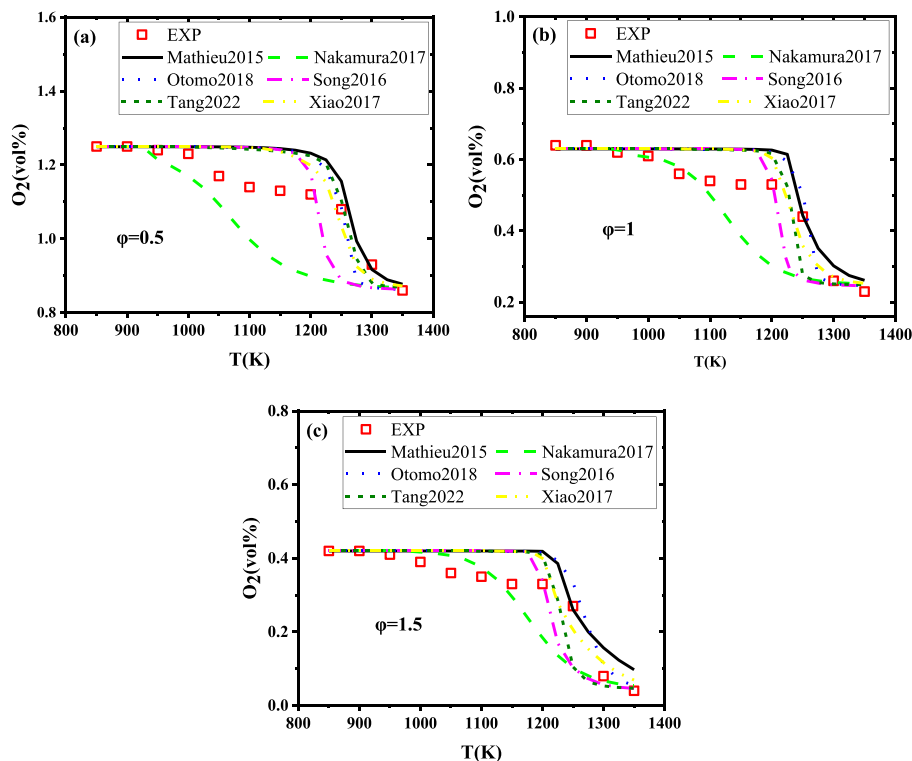


Fig. 16 – Comparisons of O_2 concentration between experimental measurement and kinetic modeling for pure NH_3 oxidation case in three equivalence ratios.

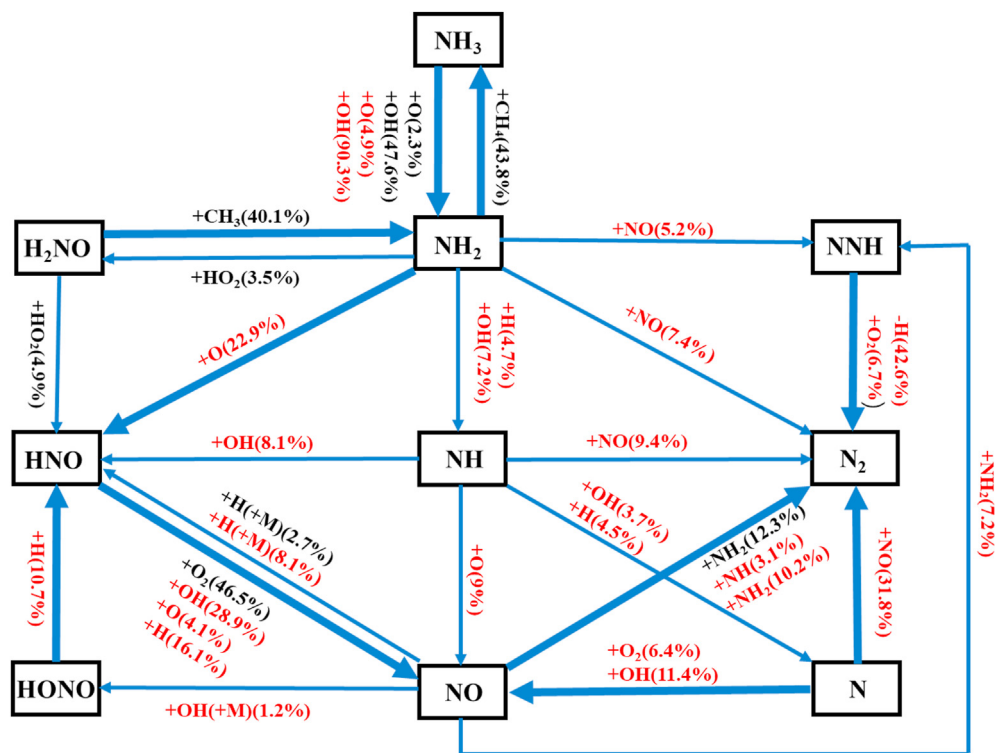


Fig. 17 – Reaction pathway of NH_3 oxidation in NH_3/CH_4 ($X_{NH_3} = 40\%$) at stoichiometric condition based on Mendiara2009 mechanism (black is 1100 K and red is 1300 K). (For interpretation of the references to color/colour in this figure legend, the reader is referred to the Web version of this article.)

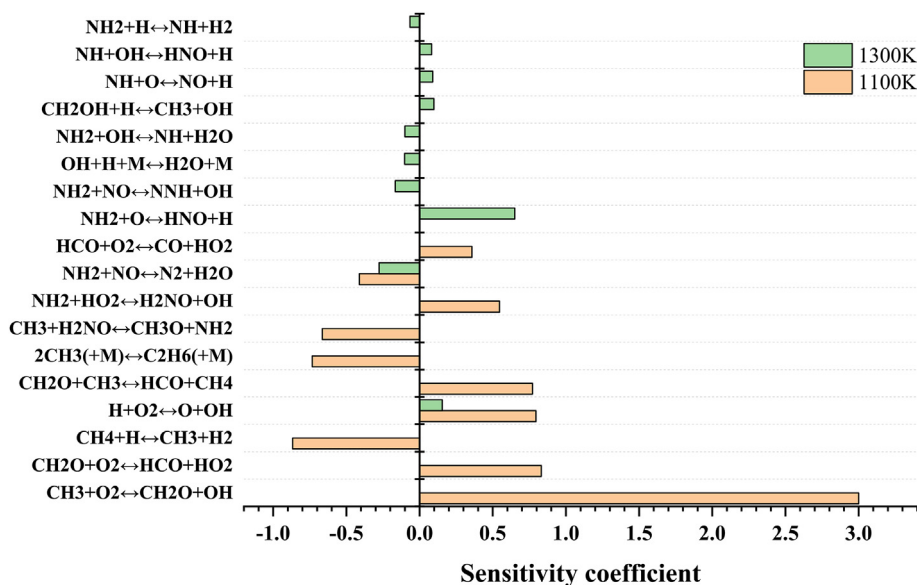


Fig. 18 – First-order sensitivity analysis of NO generation for NH_3/CH_4 ($X_{\text{NH}_3} = 40\%$) at stoichiometric condition based on Mendiara2009 mechanism.

Mendiara2009 is adopted for modeling the co-oxidation of NH_3/CH_4 mixture, while the Nakamura2017 mechanism is adopted for pure NH_3 oxidation, considering their relatively better performance in predicting NO formation in intermediate temperature (1100–1200 K) under stoichiometric condition comparing to other mechanisms. The reaction pathway analysis is conducted at two different temperatures, i.e., 1100 K and 1300 K.

Fig. 17 shows the NH_3 conversion pathway to NO for NH_3/CH_4 mixture in stoichiometric condition. For better interpretation, only the reactions with their relative contribution higher than 1% are drawn. The labels of value next to the arrow represent the relative contribution for different radicals in the corresponding elementary reaction. Furthermore, to highlight the importance of different routes, the thickness of the arrows between reactant and product is increased for

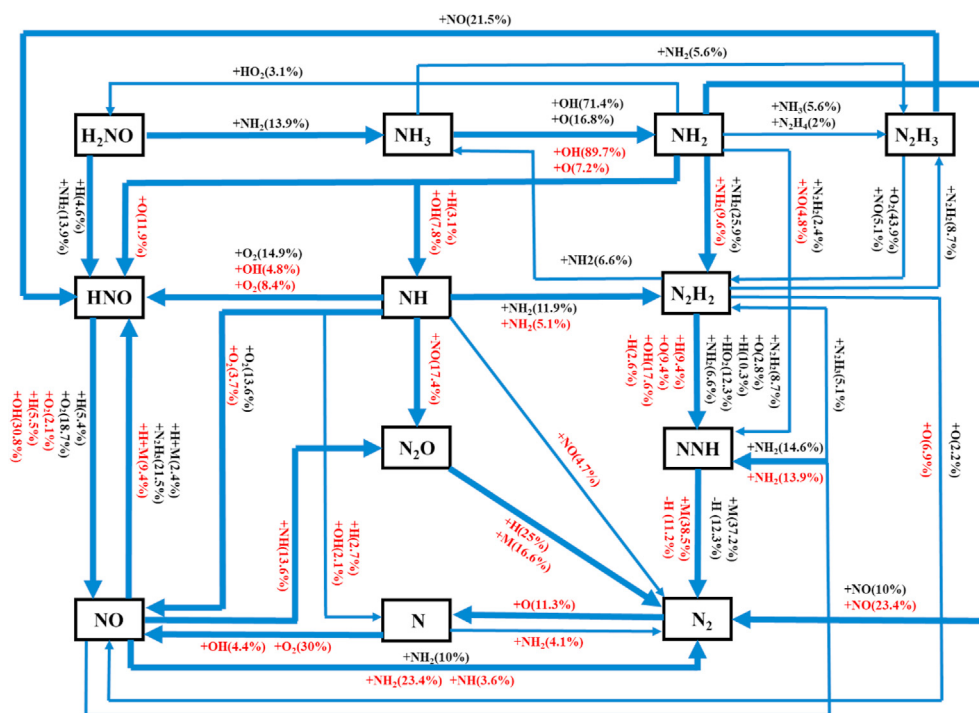


Fig. 19 – Reaction pathways of NH_3 oxidation in NH_3 at stoichiometric condition based on Nakamura2017 mechanism (black is 1100 K and red is 1300 K). (For interpretation of the references to color/colour in this figure legend, the reader is referred to the Web version of this article.)

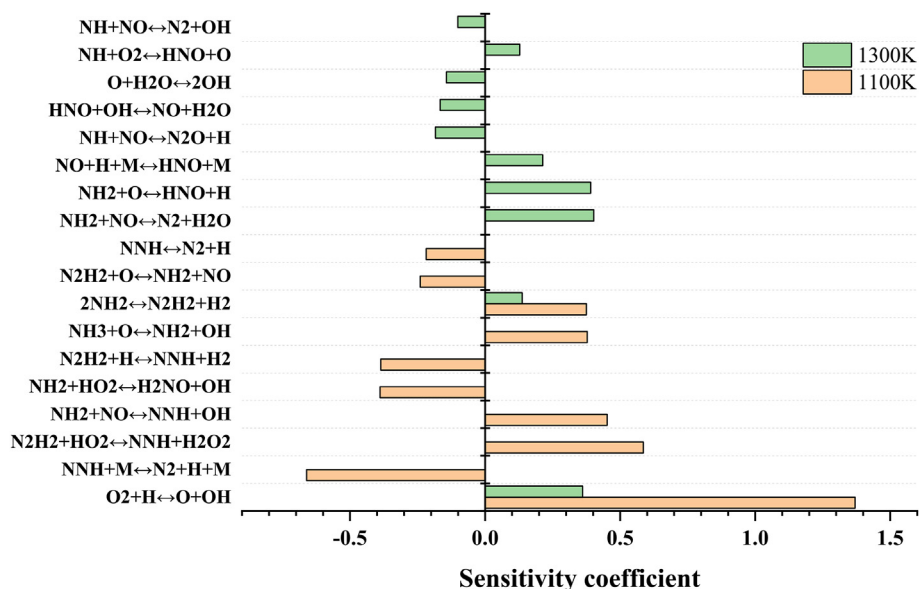


Fig. 20 – First-order sensitivity analysis of NO generation for NH_3 at stoichiometric condition based on Nakamura2017 mechanism.

those reactions with their relative contribution above 10%. The difference caused by reaction temperature is indicated by text colors, where black denotes 1100 K while red denotes 1300 K.

The reaction pathway diagram suggests that the first main conversion product from NH_3 is NH_2 via H abstraction reactions by O and OH radicals. When the temperature is boosted from 1100 K to 1300 K, the contribution of O and OH radicals for NH_3 conversion to NH_2 is raised nearly by a factor of two, while the reaction with OH radical is significantly more important than the reaction with O radical. At 1100 K, the reaction with HO_2 to produce H_2NO is the only noticeable route for NH_2 consumption. Furthermore, the CH_3 radical is found to even play an inhibition role on NH_2 consumption by converting H_2NO back to NH_2 . On the contrary, at 1300 K, NH_2 can be consumed by more routes via reactions with O, H, OH and NO. Among them HNO is the main product of NH_2 oxidation by O. Subsequently, HNO is further oxidized to NO by reaction with radicals, like OH, O and H. Hence, it is understood that the under-estimated NO formation at 1100 K by kinetic modeling is a combined result of the inhibited NH_2 oxidation and significantly strengthened NH_2 formation from H_2NO .

Fig. 18 shows the sensitivity analysis for NO production at stoichiometric and normal pressure condition with NH_3 partial pressure being 40% in the fuel mixture. The sensitivity coefficients are obtained by varying the A-factors by a factor of two for all elementary reactions in the kinetic models. It can be noticed that the reactions related to hydrocarbon radicals play a much more key role in NO formation at 1100 K, among these $\text{CH}_3 + \text{O}_2 \leftrightarrow \text{CH}_2\text{O} + \text{OH}$ has the largest sensitivity coefficient. While NO formation is basically dominated by reactions related to NH_i radicals, and $\text{NH}_2 + \text{O} \leftrightarrow \text{HNO} + \text{H}$ is of the highest sensitivity. Considering the large discrepancy between measurement and prediction in NO production at low-intermediate temperature according to Fig. 12, it is expected

to make further optimization for reactions in the subsets of H_2NO , NH_2 , and C–N interaction in the mechanism of Mendiara2009, in order to obtain improved NO prediction at this temperature range.

Fig. 19 draws the NH_3 reaction pathway for pure NH_3 oxidation at stoichiometric condition with the mechanism of Nakamura2017. NH_2 is found likewise to be the main product for NH_3 consumption, while the difference in the relative contribution from OH oxidation between 1100 K and 1300 K is not as large as that in NH_3/CH_4 mixture according to Fig. 17. More noticeably, the relative contribution from O oxidation is even reduced at the higher reaction temperature, which is opposite to the observation for NH_3/CH_4 mixture. At 1100 K, the recombination with NH_2 and NO to form N_2H_2 and N_2 , respectively, is the two main routes for NH_2 consumption. While at 1300 K, NH_2 is mainly consumed by reaction with NO to form N_2 , and additionally with O to form HNO. Simultaneously, the conversion to NH and N_2H_2 also contribute to NH_2 consumption. Basically, the consumption of NH_3 happens in four routes, i.e., (1) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$; (2) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{N}_2$; (3) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{NH} \rightarrow \text{N}_2\text{O} \rightarrow \text{N}_2$; and (4) $\text{NH}_3 \rightarrow \text{NH}_2 \rightarrow \text{HNO} \rightarrow \text{NO} \rightarrow \text{N}_2$. Routes 1 and 2 are the only path for NH_2 consumption at 1100 K, while all four routes contribute to NH_3 consumption at 1300 K despite that route 1 is inhibited. For the other three routes, their relative contribution on NH_2 consumption exceeds 10%, and route 2 makes the biggest. It is also necessary to point out the conversion from H_2NO to NH_3 which is only seen at 1100 K. It can be then understood that the slow NO formation at 1100 K by kinetic modeling is a joint result of suppressed NH_2 consumption and enhanced NH_3 back-conversion.

Fig. 20 compares the NO formation sensitivity analysis results between 1100 K and 1300 K for pure NH_3 under stoichiometric and ambient pressure condition. At 1100 K, the reactions related to route 1 and 2 mentioned above, as well as the key OH generation reaction ($\text{O}_2 + \text{H} \leftrightarrow \text{O} + \text{OH}$), show a more

influencing effect on NO formation. However, at 1300 K, the reactions of high sensitivity on NO production shift to route 2 and route 4, while route 3 shows a negative effect on facilitating NO formation via $\text{NH} + \text{NO} \leftrightarrow \text{N}_2\text{O} + \text{H}$. Overall, it is suggested to make further optimization for Nakamura2017 mechanism on the elementary reactions relevant to N_2H_2 , NNH and H_2NO , to improve NO prediction in lower temperature conditions.

Conclusions

The present work reports an experimental study on the co-oxidation process for binary fuel of CH_4 and NH_3 mixture in a JSR apparatus, which features separate inlets for oxidizer and fuel. Such design avoids earlier reaction of the mixture before entering the reactor which may occur in previous designs. New experimental data is collected for CO and NO in a wide range of reaction temperature (850–1350 K), equivalence ratio (0.5–2) as a function of NH_3 mole fraction in fuel. The data is further used to examine the prediction performance of different existing kinetic models. The main conclusions can be drawn as follows:

- (1) In fuel-lean condition, there exists a strong competition between the oxidation of CH_4 and NH_3 . The addition of NH_3 in fuel mixture not only reduces the maximum CO concentration, but also elevates the reaction temperature for generating the maximum CO when NH_3 mole fraction exceeds 40%. However, the reaction temperature for maximum CO production is not changed in stoichiometric and fuel-rich conditions by NH_3 addition, and is fixed at 1200 K and 1300 K, respectively.
- (2) A dual NO formation behavior as a function of reaction temperature is observed for all CH_4/NH_3 mixtures (including pure NH_3) in no matter what equivalence ratio condition. The competition between oxidation of CH_4 and NH_3 also significantly affects the NO formation behavior in fuel-lean condition. With the addition of NH_3 in fuel, the peak NO concentration is boosted and the reaction temperature for the maximum NO generation is also elevated. However, such temperature is unchanged in stoichiometric and fuel-rich conditions with the addition of NH_3 .
- (3) The influence of equivalence ratio on CO and NO formation is strongly dependent on the reaction temperature. At a lower temperature of 1100 K, CO is formed in lean condition; while at a higher temperature of 1300 K, it is formed in rich condition. At a medium temperature of 1200 K, CO can be formed in a wide equivalence ratio range of 0.7–2.0, and the peak CO concentration is obtained in stoichiometric condition for CH_4/NH_3 mixtures with X_{NH_3} lower than 80%. In the whole range of equivalence ratio, NO emission is enhanced with NH_3 addition in fuel at the reaction temperature of 1100 K, while reduced at 1300 K. At 1200 K, the peak NO is generated at the equivalence ratio of 1.3 for all fuels of interest.
- (4) Kinetic modeling of NO formations for both CH_4/NH_3 mixture ($X_{\text{NH}_3} = 40\%$) and pure NH_3 suggests that there is still a large space for improvement of existing reaction

mechanisms on describing the oxidation processes for these fuels. Among the reaction mechanisms considered for evaluation, Mendiara2009 and Nakamura2017 show the best NO prediction in higher temperature conditions for CH_4/NH_3 mixture and pure NH_3 , respectively.

- (5) NH_3 reaction pathway and NO formation sensitivity analysis show that, both inhibited NH_2 consumption and enhanced C–N interaction cause the poor NO prediction in low-intermediate temperature range (800–1200 K) for CH_4/NH_3 mixture, while the former is the main reason for pure NH_3 . Further optimization needs to be performed on these routes for the existing kinetic models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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